A PROJECT TITLE

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Loan Default Prediction Using Random Forest and Gradient Boosting Algorithms

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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1. Abstract

Loan default prediction is a critical problem in the banking and financial services sector, where accurately identifying applicants who are likely to default enables institutions to make sound lending decisions, minimize non-performing assets, and price credit risk fairly. This project presents a machine learning based system for predicting loan default risk using two ensemble learning techniques: Random Forest and Gradient Boosting. A dataset of loan applicant records, comprising demographic details, income, credit score, existing debt obligations, employment history, and prior repayment behaviour, was used to train and evaluate both models. The dataset was preprocessed through cleaning, encoding of categorical variables, feature scaling, and a stratified train-test split to preserve the class balance between defaulters and non-defaulters. Random Forest was chosen for its robustness to noisy data and its ability to rank feature importance, while Gradient Boosting was chosen for its sequential error-correcting mechanism that often yields higher predictive accuracy on structured tabular data. Both models were trained and compared using standard classification metrics, namely accuracy, precision, recall, F1-score, and ROC-AUC, along with confusion matrices and ROC curves to visualize their discriminative performance. The results indicate that both ensemble methods achieve strong predictive performance, with Gradient Boosting showing a marginally higher ROC-AUC than Random Forest, while Random Forest offers greater interpretability through feature importance rankings. Key predictive factors identified include credit score, debt-to-income ratio, number of late payments, and prior default history. To make the system practically usable, an interactive web application was developed using Streamlit, allowing a user to input a new applicant's details and instantly receive a default-risk prediction along with a probability score from either model. This project demonstrates how ensemble machine learning techniques can be effectively applied to credit risk assessment, providing financial institutions with a transparent, data-driven decision-support tool that complements traditional credit scoring approaches.

2. Introduction

Financial institutions extend credit to individuals and businesses with the expectation of repayment over an agreed period. However, a fraction of borrowers fail to meet their repayment obligations, resulting in loan defaults that directly impact the profitability and stability of lending institutions. Traditional credit assessment methods often rely on static rule-based scoring systems that may not capture complex, non-linear relationships between an applicant's financial profile and their likelihood of default. Machine learning offers a data-driven alternative capable of learning these intricate patterns directly from historical lending data. Ensemble learning methods, in particular, have proven highly effective for structured, tabular financial data because they combine the predictions of multiple decision trees to reduce variance and bias. Random Forest builds a large number of decision trees on bootstrapped samples of the data and aggregates their predictions through majority voting, making it robust to overfitting and noisy features. Gradient Boosting, on the other hand, builds trees sequentially, with each new tree correcting the errors made by the previous ones, often resulting in superior predictive accuracy at the cost of higher computational complexity. This project applies both techniques to the loan default prediction problem, compares their performance using a comprehensive set of evaluation metrics, and packages the resulting models into an interactive glass-themed web application that allows real-time risk assessment for new loan applicants.

Objectives of the project:

- To study and preprocess loan applicant data suitable for default prediction, including handling of numerical and categorical features.
- To build and train Random Forest and Gradient Boosting classifiers for predicting loan default.
- To evaluate and compare both models using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.
- To identify the most influential features driving loan default through feature importance analysis.
- To develop an interactive web application that allows a user to input applicant details and obtain a real-time default-risk prediction.

3. Software Requirements

Programming Language: Python 3.10 or later

Web Application Framework: Streamlit

Machine Learning Library: scikit-learn (Random Forest Classifier, Gradient Boosting Classifier)

Data Handling: pandas, NumPy

Visualization: Plotly

IDE / Editor: Visual Studio Code

Version Control: Git and GitHub

Operating System: Windows 10/11 (cross-platform compatible)

Signature of the Guide **Course Instructor**